

Jimin Han

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EDUCATION

Rice University, Texas, USA

Aug 2025 – Present

B.S. in Mechanical Engineering and B.S. in Mathematics (expected May 2029), GPA: 4.0/4.0

Relevant Coursework: Mechanics/Statics, Classical Thermodynamics, Engineering Design Tools, Industrial Processing Lab, Elements of Analysis, Differential Equations, Linear Algebra

WORK EXPERIENCE

Tribomechadynamics Lab – *Undergraduate Research Assistant* – Rice University

Oct 2025 – Present

- Develop a mini split-Hopkinson pressure bar with 4 mm Ti-6Al-4V bars under Professor Matthew Brake to characterize shock-wave reflection, transmission, and energy dissipation across bolted joints, with planned Abaqus FEA validation
- Design bar components, fixtures, T-slot framing, and an acrylic safety enclosure in SolidWorks to maintain bar alignment and improve experimental safety and impact containment, while sourcing parts across vendors and redesigning the enclosure from a hinge-and-latch lid to a sliding T-slot configuration after reviewing containment failure modes
- Prepare instrumentation for the experimental phase, including photogate triggering, high-speed camera synchronization, strain gauge placement, and DIC acquisition in LabVIEW

NSF RIDE REU (1 of 10 selected) – *Research Intern* – University of Massachusetts Amherst

Jun 2026 – Aug 2026

- Cleaned and combined 120,000+ GPS and battery records from an electric bus in Kigali, Rwanda with passenger counts from CCTV footage, elevation, and weather data to predict how much energy the bus uses, under Professor Eric Gonzales
- Compared a vehicle-dynamics model against 3 machine learning models, found that a random split let the models see test data in advance and inflated accuracy, and proposed a 60-second block split that reversed which model performed best
- Presented a research poster and first-authored a paper accepted for presentation at the TRB 2027 Annual Meeting

NSF ExLENT (1 of 20 selected) – *Research Trainee* – Michigan Technological University

Feb 2026 – May 2026

- Completed six-module mechatronics curriculum in circuit controls, hydraulics, robotics, PLCs, cybersecurity, and AI
- Applied the curriculum in person at Michigan Tech: programmed FANUC robots (frames, end-of-arm tooling, motion programs), wrote Allen-Bradley ladder logic with timers and counters, and built hydraulic circuits
- Visited four regional manufacturers to see production processes firsthand, from PCB fabrication at Calumet Electronics to vacuum-chamber testing of electric propulsion at Orbion Space Technology

Lukkey AG – *Engineering Support Intern* – Remote

Dec 2025 – Jan 2026

- Assessed Korean market entry for a Swiss hardware Bitcoin wallet by comparing demand, regulation, competitors, and pricing, identifying weak brand recognition and trust as the main barrier
- Tested an early unit and reported usability and documentation feedback to the engineering team ahead of launch
- Localized the full technical terminology set into Korean and met with the team across 10 sessions to align on terms, KRW pricing, and deployment strategy

Xcellent Life – *Strategy Intern* – Nashville, TN

Jun 2025 – Jul 2025

- Analyzed regulatory, cultural, and logistical constraints across 3 healthcare-adjacent markets to guide market entry strategy with the CEO
- Built email automation templates that cut manual outreach time and drove investor contact with 10+ firms, leading to a multi-investor meeting

PROJECTS

Battle Bots – *Subteam Leader & Mechanical Engineer at Rice Robotics Club* – Rice University

Sep 2025 – Present

- Designed, built, and competed with 3 iterations of an antweight combat robot in Onshape, fabricating parts by 3D printing, waterjet, machining, and soldering
- Redesigned a chipping weapon with a wider edge and greater infill, keyed a loosening motor shaft, and lowered the center of mass to stop flipping, placing 2nd at Rice
- Lead twice-weekly build sessions for 15 members and am planning Rice's open combat robotics competition (Fall 2027), managing venue, catering, and logistics after Houston's previous host retired

Rocket Recovery – *Mechanical Engineer at Rice Eclipse* – Rice University

Sep 2025 – Feb 2026

- Collaborated with electrical and computer engineers on dual-stage test rockets that successfully exceeded Mach 2, focusing on safe and reliable recovery
- Designed recovery modules for airframe separation, pressurization, parachute deployment, and avionics mounting
- Applied OpenRocket, SolidWorks, MATLAB, and fabrication tools to iterate designs and produce launch-ready hardware

SKILLS

Design & Analysis: SolidWorks, Onshape, Engineering Drawings, GD&T, BOM, QGIS, OpenRocket, DIC, FEA

Manufacturing: Fabrication, 3D Printing, Waterjetting, Laser Cutting, Mill, Lathe, Soldering

Programming & Data: MATLAB, Python, Pandas, NumPy, Scikit-Learn, XGBoost, SHAP, Matplotlib, Excel, LaTeX, LabVIEW

Languages: Korean (native), English (fluent), Mandarin (working proficiency)

